Supplemental Materials

Toward Model-Free Dexterous Manipulation with Low-Cost Tactile Sensors: Learning to Control Normal Contact Force

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A. Analysis of the Chiral Problem

In the paper, we introduced the chiral problem, which flips the sign of $\dot{f}_n$ at different contact regions. Without further analysis, we could throw this problem to machine learning. However, it is crucial to understand if the contact region information can be inferred at all. Otherwise, the learning may be destined to failure. Here, we take a closer look into this problem.

What the learned policy regulates is essentially the relative normal speed between the two contacting surfaces. Let us denote the relative normal speed as v_{rel} and the normal velocity offset at the contact as v_{ofs} (note that this is a linear velocity offset resulted from a_{ofs}).

Fig. 1 enumerates all possible combinations of contact regions and relative velocity directions. If the robot knows $|v_{rel}|$, it can always generate a v_{ofs} bigger than $|v_{rel}|$ and check the sign of $\dot{f}_n$. In this case, if v_{ofs} and $\dot{f}_n$ have the same sign, the contact is in one region; if the sign is different, the contact is for sure in the other region. However, v_{rel} is not a measured value (and is difficult to measure). The policy can only observe its immediate previous action and the resulting force change, $\dot{f}_n$, not v_{rel} or $|v_{rel}|$.

Now depending on the relationship between v_{ofs} and $|v_{rel}|$, we can divide the scenarios into three segments for each of the four contact cases. The segments are listed below each diagram. In each table, the first row states the three-scenario segments. The second row reflects the sign of $\dot{f}_n$ resulting from the action. For each segment, the sign of $\dot{f}_n$ is deterministic, and we may use the sign of v_{ofs} and $\dot{f}_n$ as a feature to infer the contact region. The third row is the extracted feature, $sign(\dot{f}_n v_{ofs})$. For the first and third scenario segments, the feature states the contact is in the left region if it is -1 and the right region if 1. This conflicts with the second scenario segment, which states the opposite. Thus, the controller cannot distinguish the contact regions with the inputs used in the examples.

In our implementation, v_{ofs} is not observable. However, the above conclusion still holds if we use a_{ofs} to replace v_{ofs} in the feature, since $sign(v_{ofs}) = sign(a_{ofs})$.

Following the same analysis, we noticed that when the $\ddot{f}_n$ is also given, the contact region can be determined. However, this observation is noisy in simulation, probably due to integration errors, and especially noisy in the hardware robot system when the communication latency and noise come into play. Instead of relying on the $\ddot{f}_n$, an alternative is to let the policy observe a sequence of previous action-response pairs. This would also be useful if the policy learned "probing" actions that may embed information in the sequence.

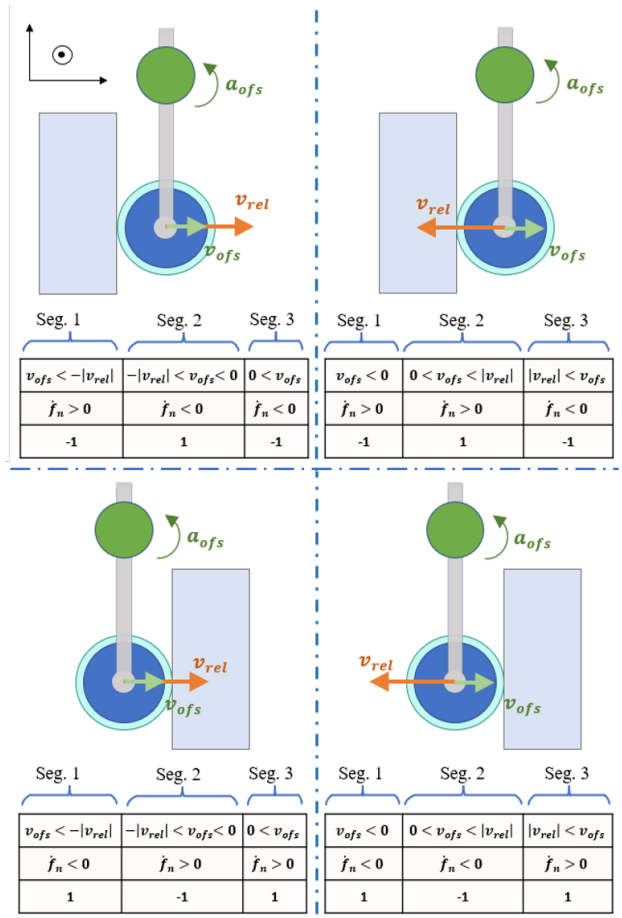

Fig. 1. All scenarios of contact regions and relative velocity directions.

B. More Details on Benchmarks

1) The choice of T in Benchmark 1: The variable T in the mean absolute off-set determines how many timesteps we measure after the initial change of the target force, and

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the choice of T affects the reported controller performance. Let us look at a typical control plot of the learned policy in Fig. 2.

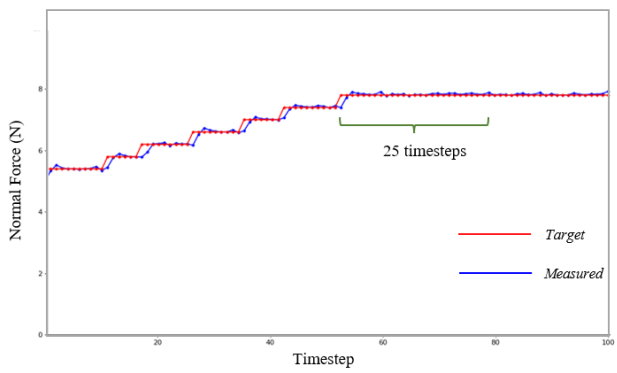

Fig. 2. Typical control measurements during force increments for the learned policy.

When the step-change in target force happens, there can be a delay in response from the controllers, and the rise time following that is non-negligible. We want a fast-responding controller, but speed may come with a cost - overshooting. From our observations, the initial transients before reaching steady-state are often within 5-10 timesteps. Increasing T will start to blend in the steady-state performances, and further increment of T will bury the initial transient information. Based on this observation, we think $T = 25$ is a balanced choice maintaining information for both the initial transients and the steady-state.

2) Benchmark Hardware Setup: We performed the two benchmarks on the hardware robotic system with the learned controller running on joint 6 at 5Hz. The force control frequency is limited by the latency of the GEN3 robot, which is around 30ms despite the advertised 1kHz robot control frequency. Based on Assumption 2, a force control timestep of 200ms is chosen to be much larger than the latency.

In both benchmarks, the tactile sensor was pressed against a heavy subwoofer, designed to be vibration-resistant and very stiff. The contact region is padded with a thin foam tape (same as the black stripe on the rectangular block in Fig. 1 of the paper). Moreover, the contact angle is guided by a precisely machined angle finder. For Benchmark 2, we positioned the robot in a non-singular configuration and computed the joint velocities corresponding to the disturbance velocity using pseudo-inverse of the robot Jacobian.